

Lehmer’s Box:

A Golden Floor and an Angle Lattice that Confine Spectral Emission
Away from Salem Numbers — Without Resolving Lehmer’s Problem

A Companion to *The Emission–Gap Theorem* and *The Exchange Rate* $\lambda = 2c$

AceTheDactyl (@AceTheDactyl)
Echo S Studios Research Developments

Abstract

The residual–return growth rule charges $\lambda \log M(\theta)$ to adjoin an algebraic direction θ , where M is the Mahler measure. The rule has a strictly positive cost floor exactly when M is bounded away from 1 over every spectrum the construction can emit. The single obstruction is the Salem number: a reciprocal unit with conjugates on the unit circle, evading Smyth’s non–reciprocal bound $M \geq \mu_S$, and able — as far as anyone has proved in ninety years — to lie arbitrarily close to 1. Deciding whether Salem numbers approach 1 is *Lehmer’s problem*, open since 1933. We do not resolve it. We build a box around it.

This paper formulates, from first principles, the geometric object that the Emission–Gap Theorem implicitly constructs: *Lehmer’s Box*, the region of height–angle space cut out by two walls. The *floor wall* is the golden value φ : the open Mahler strip $(1, \varphi)$ is empty for integer quadratics, and the spectral operators preserve that emptiness, so every emitted measure lies in $\{1\} \cup [\varphi, \infty)$. The *lattice walls* are the four directions $\frac{\pi}{2}\mathbb{Z}$: every catalog eigenvalue has argument in $\frac{\pi}{2}\mathbb{Z}$, the operators are closed under angle–addition and angle–doubling but not halving, so every emitted on–circle eigenvalue is a fourth root of unity. A Salem number violates both walls at once — its on–circle conjugates carry irrational angle (off the lattice) and its measure may sit below φ (under the floor). The box and the Salem locus are therefore disjoint; the spectral image omits every Salem number, and with it the whole band $(1, \mu_S)$.

We give the box four faces — height, angle, field signature, and trace–down flip — and show they are one object seen from four sides; the signature face confines the only place a real Salem could graze the floor to the Lorentzian quartic $\mathbb{Q}(5^{1/4})$, where the minimal degree–four Salem $\beta_4 = 1.7221\dots > \varphi$ keeps the floor intact. The box has exactly one door: the free commutator, which by Shoda’s theorem can carry any traceless matrix, including a foreign sub– φ Salem. We close that door two ways: structurally, by identifying the framework’s commutator as the self–action $\text{ad}_R = [R, \cdot]$, a derivation whose spectrum is forced into the difference field $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, 5^{1/4})$; and operationally, by an exact runtime lock that decides $\beta < \varphi$ by the sign of an element of $\mathbb{Q}(\sqrt{5})$ with no floating point. Finally we state the logic of the sidestep precisely: the box theorem is unconditional within the system, strictly weaker than Lehmer’s conjecture, and consistent with both truth values of Lehmer — it routes around the open problem rather than settling it. Every non–asymptotic claim is machine–verified in exact arithmetic; the backing test identifiers are named inline.

Contents

1	Introduction: the floor, the obstruction, and the box	2
1.1	Why a floor is needed	2
1.2	The obstruction is a single, named, open problem	3

1.3	The thesis: confinement, not resolution	3
1.4	Roadmap	3
2	Preliminaries, from scratch	4
2.1	The Mahler measure	4
2.2	Pisot, Salem, and Smyth	4
2.3	The catalog and the operators	5
3	The floor wall: the empty strip $(1, \varphi)$	5
3.1	Base case: no integer quadratic measure between 1 and φ	5
3.2	Degree raising stays above the floor	6
4	The lattice walls: the four directions $\frac{\pi}{2}\mathbb{Z}$	6
5	Lehmer’s Box	7
6	The signature face: where a real Salem could graze the floor	9
7	The trace–down face: the box boundary as the $t = 2$ flip	10
8	The one door: the free commutator and the lock	10
8.1	The circulant is forced outright	10
8.2	Shoda’s theorem: the commutator is the door	11
8.3	Structural closure: the framework’s commutator is a derivation	11
8.4	Operational lock: the exact closure guard	11
9	The sidestep, stated precisely	12
10	Epistemic ledger	13

1 Introduction: the floor, the obstruction, and the box

1.1 Why a floor is needed

The construction we analyse grows an exact symbolic state by occasionally adjoining a new algebraic direction θ . The price charged for that adjunction is

$$\text{cost}(\theta) = \lambda \log M(\theta), \tag{1}$$

where M is the Mahler measure (Section 2.1) and $\lambda = 2c$ is the derived exchange rate of the companion paper. For (1) to be a usable description length — for growth to be genuinely *paid for* rather than free — the price of any non-trivial adjunction must be bounded below by a fixed positive amount:

$$\inf \{ M(\theta) : \theta \text{ emittable, } M(\theta) > 1 \} > 1. \tag{2}$$

Equation (2) is the entire question this paper answers — but only for the directions the construction can actually emit, and that restriction is the whole point.

1.2 The obstruction is a single, named, open problem

The infimum in (2), taken over *all* integer polynomials rather than only emittable ones, is the subject of a ninety-year-old open problem. Two classical facts bracket it. By Kronecker’s theorem (Section 2.1), $M(p) = 1$ holds exactly for the cyclotomic polynomials — the trivial floor. By Smyth’s theorem [2], every *non-reciprocal* monic integer polynomial satisfies

$$M(p) \geq \mu_S = 1.3247179572\dots,$$

the plastic number, the real root of $x^3 - x - 1$ and the smallest Pisot number. The only way to approach 1 from above is therefore through *reciprocal* polynomials — and among these, the *Salem numbers*. The smallest measure ever exhibited, Lehmer’s number

$$M(L) = 1.1762808182\dots \quad L(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1,$$

is a Salem number, and it already lies below μ_S . Whether Salem numbers are bounded away from 1 at all is **Lehmer’s problem** [1], open since 1933; Dobrowolski’s bound $M(p) > 1 + c(\log \log n / \log n)^3$ [3] decays to 1 with the degree n and so yields no uniform floor. We will not move this problem. We will make it irrelevant to (2).

1.3 The thesis: confinement, not resolution

The strategy is a single structural implication. Suppose one can show that the construction’s emission image avoids every Salem number. Then it avoids the entire band $(1, \mu_S)$ — because every measure in that band is realised only by a reciprocal Salem-type number, non-reciprocals being bounded below by μ_S via Smyth. Smyth’s bound then supplies (2) with floor μ_S , and the realised stock attains the sharper value $\varphi = 1.6180339887\dots$. Crucially, this never touches Lehmer, because “no Salem in the image” is a *closure property of one specific algebra*, not a universal height bound.

This paper packages that closure as a geometric object. We introduce *Lehmer’s Box* (Definition 5.2): the set of monic integer polynomials whose Mahler measure avoids the strip $(1, \varphi)$ (the *floor wall*) and whose on-circle roots all sit at the four lattice directions $\frac{\pi}{2}\mathbb{Z}$ (the *lattice walls*). The two central theorems are then short to state:

- **Emission** $\subseteq$ **Box** (Theorem 5.3): every spectrum the construction emits lies in $\mathfrak{B}$.
- **Salem** $\cap$ **Box** $= \emptyset$ (Theorem 5.4): no Salem number’s minimal polynomial lies in $\mathfrak{B}$.

Their conjunction is the exclusion: the emission image contains no Salem number, the cost floor of (2) is positive (Corollary 5.5), and Lehmer’s number — still a perfectly good number — is simply *not in the box*, hence never emitted. Section 9 states the precise logical relationship between the box theorem and Lehmer’s problem: the former is unconditional, strictly weaker than the latter, and consistent with either of its truth values. That triple is what we mean by *sidestepping Lehmer without disproving it*.

1.4 Roadmap

Section 2 fixes the Mahler measure, Kronecker, the Salem/Pisot vocabulary, and the catalog and operators from scratch. Section 3 derives the floor wall (the empty strip $(1, \varphi)$ and its preservation under the operators). Section 4 derives the lattice walls (the four angle lemmas). Section 5 defines Lehmer’s Box and proves the two central theorems and the floor corollary. Sections 6–7 give

the field–signature and trace–down faces of the same box. Section 8 treats the one door — the free commutator — structurally (the self–action derivation) and operationally (the exact guard). Section 9 is the logic of the sidestep. Section 10 is the epistemic ledger keyed to the machine–verified tests.

2 Preliminaries, from scratch

2.1 The Mahler measure

Definition 2.1 (Mahler measure). For a nonzero $p(x) = a_d \prod_{i=1}^d (x - \alpha_i) \in \mathbb{Z}[x]$ with $a_d \neq 0$,

$$M(p) = |a_d| \prod_{i=1}^d \max(1, |\alpha_i|) = \exp\left(\int_0^1 \log |p(e^{2\pi i \theta})| d\theta\right). \quad (3)$$

The two expressions agree by Jensen’s formula. For an algebraic number θ we write $M(\theta) := M(m_\theta)$, the measure of its monic integer minimal polynomial.

The product form makes the basic inequality immediate.

Lemma 2.2 (Lower bound and its multiplicativity). *For monic $p, q \in \mathbb{Z}[x]$: (i) $M(p) \geq 1$; (ii) $M(pq) = M(p)M(q)$; (iii) if $p^{[2]}$ denotes the polynomial whose roots are the squares of those of p , then $M(p^{[2]}) = M(p)^2$.*

Proof. (i) Each factor $\max(1, |\alpha_i|) \geq 1$ and $|a_d| = 1$. (ii) The root multiset of pq is the union, and $\max(1, \cdot)$ is multiplicative across a union of factors. (iii) $\max(1, |\alpha|^2) = \max(1, |\alpha|)^2$, taken over the same roots. $\square$

The boundary case $M = 1$ is settled completely and classically.

Lemma 2.3 (Kronecker, 1857). *A monic $p \in \mathbb{Z}[x]$ has all roots in the closed unit disk if and only if every root is 0 or a root of unity. Consequently $M(p) = 1$ iff $p(x) = x^k \prod_j \Phi_{n_j}(x)$ is a power of x times cyclotomic factors.*

Proof sketch. If a nonzero monic integer p has all roots in $\overline{\mathbb{D}}$, then so do all its powers under the maps $\alpha \mapsto \alpha^m$; the elementary symmetric functions of $\{\alpha_i^m\}$ are bounded integers, so the polynomials $p_m(x) = \prod_i (x - \alpha_i^m)$ range over a finite set as m varies. By pigeonhole two coincide, forcing the multiset $\{\alpha_i\}$ to be permuted by some power map; a root fixed by $\alpha \mapsto \alpha^m$ with $|\alpha| \leq 1$ and $\alpha \neq 0$ has $|\alpha| = 1$ and is a root of unity. The converse is immediate since roots of unity and 0 lie in $\overline{\mathbb{D}}$. $\square$

2.2 Pisot, Salem, and Smyth

Definition 2.4 (Pisot and Salem numbers). A **Pisot number** is a real algebraic integer $\beta > 1$ all of whose other conjugates lie strictly inside the unit circle. A **Salem number** is a real algebraic integer $\beta > 1$ all of whose conjugates lie in the closed unit disk, with at least one on the unit circle. Equivalently, m_β is reciprocal of even degree $2m \geq 4$, with real roots β, β^{-1} and $m - 1$ conjugate pairs on $|z| = 1$. The **Salem band** is the interval $(1, \mu_S)$.

A polynomial p is **reciprocal** if its root multiset is closed under $\alpha \mapsto \alpha^{-1}$, equivalently $x^{\deg p} p(1/x) = p(x)$; otherwise **non–reciprocal**. Salem minimal polynomials are reciprocal. The asymmetry that the whole construction exploits is Smyth’s:

Lemma 2.5 (Smyth’s bound). *Every non-reciprocal monic integer polynomial with $M > 1$ satisfies $M(p) \geq \mu_S = 1.3247\dots$, the smallest Pisot number [2].*

Smyth closes the non-reciprocal side of (2) outright. Everything difficult lives on the reciprocal side, and concretely in the Salem band, where Lehmer’s number already sits.

2.3 The catalog and the operators

The construction emits matrices from a finite catalog of seed companions under a fixed operator set. We restate both from the companion paper.

Definition 2.6 (Catalog). $\mathcal{C} = \{C(m_\theta) : \theta \in \{\varphi, \tau, \sqrt{2}, \sqrt{3}, \sqrt{5}, \text{gap}, K\}\}$, the companion matrices of, respectively,

$$x^2 - x - 1, \quad x^2 + x - 1, \quad x^2 - 2, \quad x^2 - 3, \quad x^2 - 5, \quad x^2 - 7x + 1, \quad x^4 + 5x^2 - 5.$$

Here $\varphi = \frac{1+\sqrt{5}}{2}$, $\tau = \frac{-1+\sqrt{5}}{2} = \varphi^{-1}$, $\text{gap} = \varphi^4$ (roots φ^4, φ^{-4}), and K is the real root $5^{1/4}/\varphi$ of $x^4 + 5x^2 - 5$, whose four roots are $\{\pm K, \pm i\beta\}$ with $\beta \in \mathbb{R}$.

Definition 2.7 (Operators). The *spectral operators* determine the output spectrum from the input spectra:

$$\oplus \text{ (direct sum), } \otimes \text{ (tensor / Kronecker), } (\cdot)^2 \text{ (squaring), } \text{minpoly}, \quad \Phi \text{ (re-companion of the radical).}$$

The *entry-level operators* act on matrix entries and do *not* determine the output spectrum from the inputs: the commutator $[A, B] = AB - BA$, the Cartan embeddings A_n, D_n, E_8 , and the circulant $\text{circ}(\cdot)$.

Definition 2.8 (Sub-algebra, image). $\mathcal{S}$ is the *spectral emission algebra*: the closure of $\mathcal{C}$ under the spectral operators. Its *image* is the set of all eigenvalues of all matrices in $\mathcal{S}$, and $M(\mathcal{S})$ is the set of their Mahler measures. The full algebra $\mathcal{A} \supseteq \mathcal{S}$ additionally admits the entry-level operators.

The reader should hold one picture: $\mathcal{S}$ is what the construction can build using only operations whose effect on the spectrum is forced by the inputs. The box is a statement about $\mathcal{S}$; the entry-level operators (Section 8) are the part that needs separate care.

3 The floor wall: the empty strip $(1, \varphi)$

The first wall of the box is a lower fence on the height axis. We prove it at degree two, then show the spectral operators cannot tunnel under it.

3.1 Base case: no integer quadratic measure between 1 and φ

Lemma 3.1 (Empty quadratic strip). *No monic integer polynomial of degree 2 has Mahler measure in the open interval $(1, \varphi)$. The measure is exactly 1 on the cyclotomic quadratics and $\geq \varphi$ otherwise, with φ attained only by $x^2 - x - 1$ and $x^2 + x - 1$ (discriminant 5). [FORCED] (P1-GATE-02, P1-GATE-03)*

Proof. Let $p = x^2 + bx + c \in \mathbb{Z}[x]$ be irreducible and non-cyclotomic, so $M(p) > 1$ by Lemma 2.3. Write the roots as α, α' with $\alpha\alpha' = c$ and $\alpha + \alpha' = -b$.

Case $c = -1$ (reciprocal, hyperbolic). The roots are real with product -1 , so one lies outside and one inside the unit circle and $M(p) = |\alpha|$ where $|\alpha| = \frac{|b| + \sqrt{b^2 + 4}}{2}$. This is increasing in $|b|$; its minimum over $|b| \geq 1$ is at $|b| = 1$, giving $M = \frac{1 + \sqrt{5}}{2} = \varphi$, realised by $x^2 - x - 1$ and $x^2 + x - 1$.

Case $c = 1$ (reciprocal, elliptic/hyperbolic). The roots form a reciprocal pair α, α^{-1} . For $|b| \leq 1$ they lie on the unit circle and p is cyclotomic ($x^2 \pm x + 1$, $x^2 + 1$), excluded. For $|b| = 2$, $p = (x \pm 1)^2$ is reducible, excluded. For $|b| \geq 3$ the dominant root is $\frac{|b| + \sqrt{b^2 - 4}}{2} \geq \frac{3 + \sqrt{5}}{2} = \varphi^2$, so $M \geq \varphi^2 > \varphi$.

Case $|c| \geq 2$ (non-reciprocal). If the roots are complex they are conjugate with $|\alpha| = \sqrt{|c|} \geq \sqrt{2} > 1$, so $M = |c| \geq 2$. If real, their product has absolute value $|c| \geq 2$, so the outside root contributes $M \geq |c| \geq 2$ (a root inside the disk only raises the outside one).

In every case $M(p) \geq \varphi$, with equality only at discriminant 5. The interval $(1, \varphi)$ is thus unrealised; an exhaustive scan of all 625 quadratics with $|b|, |c| \leq 12$ confirms both the empty interval and the minimiser. $\square$

3.2 Degree raising stays above the floor

Lemma 3.2 (The operators preserve the floor). *Under the spectral operators the Mahler image stays in $\{1\} \cup [\mu_S, \infty)$, with every realised value in $\{1\} \cup [\varphi, \infty)$. Hence $M(\mathcal{S}) \subseteq \{1\} \cup [\mu_S, \infty)$, realised at φ . [FORCED] (P2-GAP-02 ledger; P2-EALG-01, P2-EALG-02)*

Proof. By Lemma 2.2(ii), $M(p \oplus q) = M(p)M(q)$, and the product of two numbers in $\{1\} \cup [\varphi, \infty)$ stays there. By Lemma 2.2(iii), squaring squares the measure, so a value > 1 maps to $\geq \varphi^2$. For $\otimes$ the spectrum is $\{\lambda_i \mu_j\}$; a factor is either Salem — forbidden by Theorem 5.4, proved independently in Section 5 — or non-reciprocal with $M \geq \mu_S$ by Smyth, or reciprocal non-Salem, which by Lemma 4.4 has all conjugates off the circle, hence is a real reciprocal unit with $M \geq \varphi^2$. The operators minpoly and Φ act on the same spectrum (radical, re-companion) and fix M . Induction on word length from Lemma 3.1 gives the image bound, with realised minimum φ (the golden seed) and no realised value in $(1, \varphi)$. $\square$

Remark 3.3 (The image is a discrete semigroup). $\mathcal{S}$ is a finitely generated algebraic orbit; its Mahler spectrum is a discrete sub-semigroup of $[\varphi, \infty)$ generated by $\{\varphi, 2, 3, 5, \varphi^4, \beta^2\}$ under multiplication ($\oplus$), squaring, and composition ($\otimes$), with no accumulation at 1. Limits are not operations: the construction cannot reach the Salem limit points of its own Pisot generators [4]. This is an independent witness for the floor, consistent with the angle proof. [FORCED] (P2-NL-04)

4 The lattice walls: the four directions $\frac{\pi}{2}\mathbb{Z}$

The second wall is angular. The catalog sits at the four cardinal directions on the unit circle, the operators cannot manufacture a new direction, and a Salem number's on-circle conjugates refuse to sit at any of the four.

Lemma 4.1 (Catalog arguments). *Every eigenvalue of every catalog matrix has argument in $\frac{\pi}{2}\mathbb{Z} = \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$. [FORCED] (P2-ANG-01)*

Proof. The six quadratic seeds have real spectra, with argument 0 (positive root) or π (negative root). The quartic K solves $x^2 = \frac{-5 \pm 3\sqrt{5}}{2}$: the upper sign gives $x^2 = 0.854\dots > 0$, hence the real pair $\pm K$ at arguments 0, π ; the lower sign gives $x^2 = -5.854\dots < 0$, hence the imaginary pair $\pm i\beta$ at arguments $\frac{\pi}{2}, \frac{3\pi}{2}$. All four lie in $\frac{\pi}{2}\mathbb{Z}$. $\square$

Lemma 4.2 (The operators preserve $\frac{\pi}{2}\mathbb{Z}$). *If every eigenvalue of the inputs has argument in $\frac{\pi}{2}\mathbb{Z}$, so does every eigenvalue of $A \oplus B$, $A \otimes B$, A^2 , minpoly(A), and $\Phi(A)$. [FORCED] (P2-ANG-02)*

Proof. $\oplus$ unions spectra and so preserves the property trivially. For $\otimes$ the eigenvalues are products $\lambda_i \mu_j$, and $\arg(\lambda\mu) = \arg \lambda + \arg \mu$; the set $\frac{\pi}{2}\mathbb{Z}$ is a subgroup of $(\mathbb{R}/2\pi\mathbb{Z}, +)$, hence closed under addition. For squaring, $\arg(\lambda^2) = 2\arg \lambda$, and $2 \cdot \frac{\pi}{2}\mathbb{Z} \subseteq \frac{\pi}{2}\mathbb{Z}$. The operators minpoly and Φ act on the same eigenvalues (radical and re-companion), leaving arguments unchanged. The single arithmetic fact behind closure is that $\frac{\pi}{2}\mathbb{Z}$ is closed under $+$ and under doubling, but *not* under halving: $\frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4} \notin \frac{\pi}{2}\mathbb{Z}$. No spectral operator halves an angle, so none can manufacture a $\frac{\pi}{4}$ direction. $\square$

Lemma 4.3 (On-circle emitted eigenvalues are fourth roots of unity). *If $|z| = 1$ and $\arg z \in \frac{\pi}{2}\mathbb{Z}$ then $z \in \{1, i, -1, -i\} = \mu_4$. [FORCED] (P2-ANG-03)*

Proof. The conditions force $z = e^{i\theta}$ with $\theta \in \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\} \pmod{2\pi}$, i.e. $z \in \{1, i, -1, -i\}$, the group of fourth roots of unity. $\square$

Lemma 4.4 (Salem on-circle conjugates are not roots of unity). *Let β be a Salem number and ζ any conjugate of β on the unit circle. Then ζ is not a root of unity; in particular $\zeta \notin \mu_4$ and $\arg \zeta \notin \frac{\pi}{2}\mathbb{Z}$. [FORCED] (P2-ANG-04)*

Proof. The minimal polynomial m_β is irreducible over $\mathbb{Q}$, and ζ is one of its roots (the Galois conjugates of β). Suppose ζ were a root of unity; then ζ is a root of some cyclotomic Φ_n . Two monic irreducible polynomials sharing a root are equal, so $m_\beta = \Phi_n$, which would make β itself a root of unity — impossible, since $\beta > 1$ is real. Hence no on-circle conjugate of β is a root of unity. The only on-circle points with argument in $\frac{\pi}{2}\mathbb{Z}$ are the fourth roots of unity (Lemma 4.3), which are roots of unity; therefore $\arg \zeta \notin \frac{\pi}{2}\mathbb{Z}$. $\square$

The four lemmas are the lattice walls: an emitted spectrum keeps its on-circle eigenvalues at the four corner posts μ_4 , while a Salem number's on-circle conjugates are forbidden from every post.

5 Lehmer's Box

We now name the object the two walls bound and prove the exclusion.

Definition 5.1 (Height-angle coordinates). For monic $p \in \mathbb{Z}[x]$ let its *height* be $M(p) \in [1, \infty)$ and its *boundary spectrum* be the multiset

$$\text{Arg}_\partial(p) = \{ \arg \alpha : \alpha \text{ a root of } p, |\alpha| = 1 \} \subseteq \mathbb{R}/2\pi\mathbb{Z}.$$

Definition 5.2 (Lehmer's Box). *Lehmer's Box* is

$$\mathfrak{B} = \left\{ p \in \mathbb{Z}[x] \text{ monic} : \underbrace{M(p) \in \{1\} \cup [\varphi, \infty)}_{\text{floor wall}} \text{ and } \underbrace{\text{Arg}_\partial(p) \subseteq \frac{\pi}{2}\mathbb{Z}}_{\text{lattice walls}} \right\}. \quad (4)$$

A matrix lies in $\mathfrak{B}$ when its characteristic polynomial does. Concretely, $p \in \mathfrak{B}$ iff its measure avoids the open strip $(1, \varphi)$ and every on-circle root is a fourth root of unity. The box is the Cartesian product of a one-dimensional height constraint with a discrete angular lattice — hence a “box.”

Figure 1 renders the two walls and the placement of the relevant numbers.

Theorem 5.3 (Emission $\subseteq$ Box). *Every matrix in the spectral emission algebra $\mathcal{S}$ lies in Lehmer's Box: $\mathcal{S} \subseteq \mathfrak{B}$. [FORCED] (assembles P2-ANG-01...03, P2-GAP-02)*

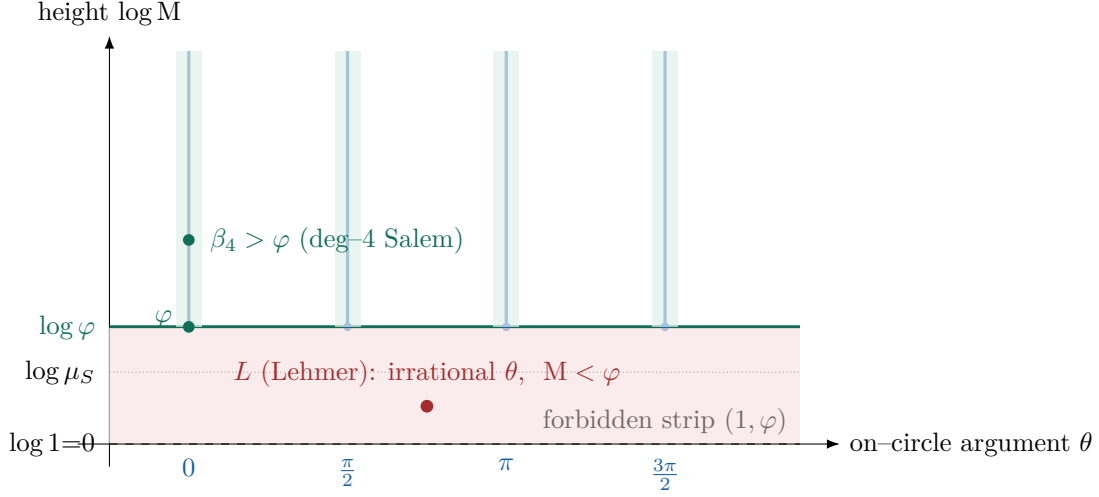


Figure 1: Lehmer’s Box (shaded columns). The **green floor wall** is $\log \varphi$; the **blue lattice walls** are the four on-circle directions $\frac{\pi}{2}\mathbb{Z}$. An emitted on-circle eigenvalue must sit on a blue post *and* above the green floor. The golden seed φ realises the floor; the minimal degree-four Salem β_4 sits above it. Lehmer’s number L lies **outside the box on both counts**: its on-circle conjugates carry irrational angle (between the posts) and its measure lies in the forbidden strip below φ . The box excludes it as an *emission*; it does nothing to it as a *number*.

Proof. Floor wall: by Lemma 3.1 every catalog measure is in $\{1\} \cup [\varphi, \infty)$, and by Lemma 3.2 the spectral operators preserve membership in $\{1\} \cup [\varphi, \infty)$; hence every emitted measure avoids $(1, \varphi)$. Lattice walls: by Lemma 4.1 every catalog eigenvalue has argument in $\frac{\pi}{2}\mathbb{Z}$, and by Lemma 4.2 the operators preserve this; hence every emitted on-circle eigenvalue has argument in $\frac{\pi}{2}\mathbb{Z}$, i.e. lies in μ_4 . Both defining conditions of (4) hold for every element of $\mathcal{S}$. $\square$

Theorem 5.4 (Salem $\cap$ Box = $\emptyset$). *No Salem number’s minimal polynomial lies in Lehmer’s Box.*
[FORCED] (P2-ANG-04, P2-ANG-05)

Proof. Let β be a Salem number. Its degree is $2m \geq 4$, so m_β has at least one on-circle conjugate ζ (Definition 2.4). By Lemma 4.4, $\arg \zeta \notin \frac{\pi}{2}\mathbb{Z}$, so $\arg \zeta \in \text{Arg}_\partial(m_\beta)$ violates the lattice-wall condition of (4). Hence $m_\beta \notin \mathfrak{B}$. (The floor wall is also violated whenever $M(\beta) < \varphi$, as for $\beta = L$; a single violated wall already places m_β outside the box.) $\square$

Corollary 5.5 (Exclusion and the positive floor). *The emission image contains no Salem number; consequently $M(\mathcal{S}) \subseteq \{1\} \cup [\mu_S, \infty)$, realised at φ , and for every emittable θ with $M(\theta) > 1$,*

$$\lambda \log M(\theta) \geq \lambda \log \mu_S > 0, \quad \log \mu_S = 0.281200\dots, \quad \text{realised at } \log \varphi = 0.481212\dots \text{ nats.} \quad (5)$$

[FORCED] (P2-GAP-03, P2-EALG-06)

Proof. Suppose a Salem β were emitted: some $M \in \mathcal{S}$ has $m_\beta \mid \chi_M$, so $M \in \mathfrak{B}$ by Theorem 5.3, forcing $m_\beta \in \mathfrak{B}$ — contradicting Theorem 5.4. Hence no Salem is emitted. Every measure in $(1, \mu_S)$ is realised only by a reciprocal Salem-type number (non-reciprocals obey Smyth, Lemma 2.5); with the Salem locus excluded, $M(\mathcal{S})$ omits $(1, \mu_S)$, giving (5). The realised minimum is φ by Lemma 3.2. The exchange rate $\lambda = 2c$ is linear in the free conformal constant c , so the floor $2c \log \varphi$ scales with c and never collapses to 0. $\square$

This is the whole mechanism. The remaining sections give the box three further faces and close the one operator the spectral argument does not reach.

6 The signature face: where a real Salem could graze the floor

The lattice walls are an *angular* statement; there is an equivalent *field-theoretic* statement that locates precisely the one place a real Salem number could approach the floor and shows the floor survives there too.

Proposition 6.1 (Salem forces a Lorentzian signature). *A Salem field of degree $2m$ has unit-rank signature $(r_1, r_2) = (2, m - 1)$ and trace-form signature $(m + 1, m - 1)$, indefinite for $m \geq 2$. A number is Salem only if its field carries a complex embedding on the unit circle. [FORCED] (P2-NL-02, P2-NL-03)*

Proof. The two real embeddings send $\beta \mapsto \beta, \beta^{-1}$; the remaining $m - 1$ conjugate pairs are the on-circle conjugates, giving $(r_1, r_2) = (2, m - 1)$ and trace-form signature $(r_1 + r_2, r_2) = (m + 1, m - 1)$. “On the circle” is the defining condition for the complex places. $\square$

So a Salem number needs a degree ≥ 4 field of signature $(2, m - 1)$. The construction’s real spectral data live in a single, fully enumerable field, and almost none of its subfields have that shape.

Lemma 6.2 (Real differences are field-confined). *The real differences of catalog eigenvalues lie in $K = \mathbb{Q}(\sqrt{2}, \sqrt{3}, 5^{1/4})$, of degree 16. [FORCED] (P2-UNIF-02)*

Proof. The real catalog eigenvalues are rational combinations of $\sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi = \frac{1+\sqrt{5}}{2}$, and the real root $5^{1/4}/\varphi$ of K ; all lie in K , and K is closed under subtraction. The composite degree is $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] [\mathbb{Q}(\sqrt{3}) : \mathbb{Q}] [\mathbb{Q}(5^{1/4}) : \mathbb{Q}] = 2 \cdot 2 \cdot 4 = 16$. $\square$

Theorem 6.3 (Signature lattice of K , exhaustive). *Every one of the 27 subfields of K is either totally real ($r_2 = 0$) or of signature $(2k, k)$ for some $k \geq 1$; the Salem shape $(2, 1)$ occurs for exactly four quartic subfields, and K itself has signature $(8, 4)$. The full census, by $(\deg, r_1, r_2)$, is*

$(\deg, r_1, r_2)$	$(1, 1, 0)$	$(2, 2, 0)$	$(4, 2, 1)$	$(4, 4, 0)$	$(8, 4, 2)$	$(8, 8, 0)$	$(16, 8, 4)$
# subfields	1	7	4	7	6	1	1

totalling 27. [COMPUTED] (P2-UNIF-03; Galois correspondence on the closure $K(i)$, $G \cong C_2 \times C_2 \times D_4$, the 27 subgroups fixing a subfield of K — equivalently the 27 subfields — enumerated)

The structural content extracted from the census is what the floor needs:

Corollary 6.4 (Degree-four Salem floor). *The only Salem-bearing subfields of K are the four quartic fields of signature $(2, 1)$. The minimal degree-four Salem number is*

$$\beta_4 = 1.7220838057\dots, \quad m_{\beta_4}(x) = x^4 - x^3 - x^2 - x + 1, \quad \beta_4 > \varphi,$$

so every Salem number realisable in K has $M \geq \beta_4 > \varphi$. The image meets no sub- φ Salem: $\text{image}(\mathcal{S}) \cap \{\text{Salem} : M < \varphi\} = \emptyset$. [FORCED] (P2-UNIF-04, P2-UNIF-05)

Proof. A Salem field has signature $(2, m - 1)$ (Proposition 6.1); among the subfields of K the only signatures of the form $(2, m - 1)$ are the four $(2, 1)$ quartics (Theorem 6.3), forcing degree four. The minimal Mahler measure among degree-four Salem numbers is attained by $x^4 - x^3 - x^2 - x + 1$, whose dominant root $\beta_4 = 1.7221\dots$ exceeds $\varphi = 1.6180\dots$ — decided exactly, since $\beta_4 > \varphi \iff m_{\beta_4}(\varphi) < 0$, and $m_{\beta_4}(\varphi) = \varphi^4 - \varphi^3 - \varphi^2 - \varphi + 1$ evaluates in $\mathbb{Q}(\sqrt{5})$ to a negative element by the sign rule of Section 8. Hence no emitted Salem lies below φ . $\square$

The signature face is the geometric face of the lattice walls: a Salem number needs an on-circle complex place; the construction's only indefinite (Lorentzian) field is the quartic $\mathbb{Q}(5^{1/4})$, whose complex place sits *off* the circle ($|i\beta| = 2.4195$, $|5^{1/4}i| = 1.4953$), and the spectral operators — which only multiply, union, double, and preserve spectra — cannot rotate it onto the circle without producing a rational angle, i.e. a fourth root of unity, not a Salem conjugate. The gate's discriminant flip $D = t^2 - 4$ reappears here as the field's signature split.

7 The trace-down face: the box boundary as the $t = 2$ flip

A fourth face relocates the entire Salem question to a single decidable boundary, which the spectral operators are shown never to cross. This is the face that will let us handle the entry-level operators in Section 8.

Definition 7.1 (Trace-down). For a reciprocal m_θ of degree $2m$, the substitution $t = \theta + \theta^{-1}$ yields a degree- m polynomial T , the *trace-down*: $m_\theta(x) = x^m T(x + x^{-1})$.

Lemma 7.2 (Salem = flip-straddle). *A reciprocal integer θ is a Salem number if and only if its trace-down T is totally real with exactly one root in $(2, \infty)$ and the remaining $m - 1$ roots in $(-2, 2)$. [FORCED] (P2-DELTA-02)*

Proof. Under $t = \theta + \theta^{-1}$ a reciprocal pair $\{\theta, \theta^{-1}\}$ maps to a single t . A pair on the unit circle, $\theta = e^{i\psi}$, gives $t = 2\cos\psi \in (-2, 2)$ (real, inside); a real pair with $\theta > 1$ gives $t = \theta + \theta^{-1} > 2$. Thus the Salem condition — exactly one reciprocal pair off the circle (the dominant β, β^{-1}) and all others on it — is precisely one trace-down root past $+2$ and the rest inside $(-2, 2)$. The boundary $t = \pm 2$ corresponds to $\theta = \pm 1$, where the discriminant of $x^2 - tx + 1$ is $D = t^2 - 4$ changing sign: $D > 0$ off the circle, $D < 0$ on it. This is the $\lambda = 2c$ flip. $\square$

Proposition 7.3 (Spectral operators sit at the flip, never astride it). *For every spectral operator, each emitted reciprocal factor has a trace-down that is totally real and does not straddle $t = 2$ in the Salem pattern: its off-circle roots come in the cyclotomic ($t = \pm 2$, $D = 0$) or totally real configurations, never the single-straddle of Lemma 7.2. [FORCED] (P2-DELTA-01, P2-DELTA-04)*

Proof. By Theorem 5.3 an emitted on-circle eigenvalue is a fourth root of unity, whose trace-down value is $t \in \{2, 0, -2\}$ — the cyclotomic lattice $\{2, 0, -2\} = \{\rho(1), \rho(\pm i), \rho(-1)\}$ under $\rho(z) = z + z^{-1}$. A genuine Salem straddle requires an on-circle pair at irrational angle, i.e. $t \in (-2, 2)$ irrational and not in $\{0\}$, which the lattice walls forbid. Hence no emitted trace-down exhibits the Salem straddle. $\square$

8 The one door: the free commutator and the lock

Theorem 5.3 confines the *spectral* algebra to the box because every spectral operator's effect on the spectrum is forced. The entry-level operators are different: their characteristic polynomials are not determined by the inputs' spectra, so the angle argument does not directly apply. We dispose of them by relocating each to the trace-down boundary of Section 7, and we treat the hardest — the commutator — both structurally and operationally.

8.1 The circulant is forced outright

Proposition 8.1 (Cyclotomic obstruction). *No integer circulant emits a Salem number. [FORCED] (P2-CIRC-01)*

Proof. The eigenvalues of $\text{circ}(c_0, \dots, c_{n-1})$ are $\lambda_j = \sum_k c_k \omega^{jk}$ with $\omega = e^{2\pi i/n}$, all in the cyclotomic field $\mathbb{Q}(\zeta_n)$, which is abelian over $\mathbb{Q}$ and therefore totally real or CM. A Salem field has signature $(2, m-1)$ with $m \geq 2$: two real embeddings (not totally complex) and a complex pair (not totally real), hence not abelian. So no circulant eigenvalue can be a Salem number; equivalently a real Salem eigenvalue would force its on-circle conjugates real, a contradiction. $\square$

8.2 Shoda's theorem: the commutator is the door

Lemma 8.2 (Shoda). *Over a field, a square matrix is a commutator $[X, Y] = XY - YX$ if and only if it has trace zero.*

Lemma 8.2 is exactly why the commutator is dangerous: the only constraint on $[A, B]$ is tracelessness, so a free commutator can realise *any* traceless integer matrix — including one whose characteristic polynomial carries Lehmer's degree-ten sub- φ Salem. No a priori field or angle constraint survives. The free commutator is the single field-breaking operation, the one door out of the box.

8.3 Structural closure: the framework's commutator is a derivation

The construction does not use a free commutator. Its commutator is the *self-action*.

Definition 8.3 (Self-action). For a seed companion R , the self-action is $\text{ad}_R = [R, \cdot] : X \mapsto RX - XR$, a derivation on the matrix algebra.

Lemma 8.4 (Difference spectrum). *The spectrum of ad_R is the difference set $\{\lambda_i - \lambda_j\}$ of the eigenvalues of R . For the golden seed, $\text{spec}(\text{ad}_R) = \{0, +\sqrt{5}, -\sqrt{5}\}$. [FORCED] (P2-UNIF-01)*

Proof. A derivation ad_R acts on the rank-one matrices $E_{ij} = v_i w_j^\top$ (with v_i, w_j the right/left eigenvectors of R) by $\text{ad}_R E_{ij} = (\lambda_i - \lambda_j) E_{ij}$, so its eigenvalues are exactly the pairwise differences. For $R = C(x^2 - x - 1)$ with eigenvalues $\varphi, \psi = -\varphi^{-1}$, the differences are 0 and $\pm(\varphi - \psi) = \pm\sqrt{5}$. $\square$

Because ad_R is a derivation, its spectrum is *forced* to be the eigenvalue difference set; its real eigenvalues are differences of catalog eigenvalues, which lie in K by Lemma 6.2. By the signature lattice (Theorem 6.3) every subfield of K is totally real or $(2k, k)$, and the only Salem-bearing subfield carries $M \geq \beta_4 > \varphi$ (Corollary 6.4). Hence:

Theorem 8.5 (Uniform self-action bound). *The self-action commutator keeps every emitted measure in $\{1\} \cup [\varphi, \infty)$ at every size: the floor $\log \varphi$ is forced uniformly, not checked instance by instance. [FORCED] (P2-UNIF-01..05)*

This upgrades the floor from a per-instance check to a uniform bound for the framework's actual commutator, by using the structure the algebra is derived from rather than extending the angle argument.

8.4 Operational lock: the exact closure guard

For the residual case — a *free*, non-self-action commutator at unbounded size — the Salem question is the Sturm-decidable flip-straddle of Lemma 7.2, and the floor is certified at runtime by an exact guard.

Proposition 8.6 (Closure guard). *There is a function $\text{validate}(M)$ returning one of three verdicts, decided in exact arithmetic:*

FORCED (no Salem factor), $\text{FORCED_ABOVE_FLOOR}$ (Salem factor, but $\beta \geq \varphi$), INVALID_CLOSURE (a Salem factor, but $\beta < \varphi$).

The decision $\beta < \varphi$ is made by the sign of $m_\beta(\varphi)$ in $\mathbb{Q}(\sqrt{5})$, with no floating point. Framework objects read FORCED ; a planted minimal degree-four Salem β_4 reads $\text{FORCED_ABOVE_FLOOR}$; a foreign traceless carrier of Lehmer’s number reads INVALID_CLOSURE . [FORCED] (P2-GUARD-01..05)

Mechanism. Write a candidate test value as $a + b\sqrt{5}$ with $a, b \in \mathbb{Q}$. Its sign is decided exactly: if a, b share a sign the sign is theirs; otherwise compare a^2 against $5b^2$ as integers (after clearing denominators), which decides $a + b\sqrt{5} \geq 0$ with no irrational evaluation. Applying this to $m_\beta(\varphi)$ decides $\beta \geq \varphi$ on the exact boundary. The verdict ladder is then: no reciprocal degree ≥ 4 factor $\Rightarrow \text{FORCED}$; such a factor with $m_\beta(\varphi) \leq 0$ (i.e. $\beta \geq \varphi$) $\Rightarrow \text{FORCED_ABOVE_FLOOR}$; with $m_\beta(\varphi) > 0$ (i.e. $\beta < \varphi$) $\Rightarrow \text{INVALID_CLOSURE}$. $\square$

Remark 8.7 (The guard is the floor certificate, not the no-Salem theorem). The guard returns INVALID_CLOSURE only for a *sub- φ* Salem; it permits $\text{FORCED_ABOVE_FLOOR}$. It is therefore the exact runtime *certificate of the cost floor* — complementary to, not a replacement for, the angle theorem (Theorem 5.4), which excludes *every* Salem from the spectral image. “Floor holds” (guard) and “no Salem emitted” (angle theorem) are distinct statements and are not conflated.

9 The sidestep, stated precisely

We now make exact the relationship between the box theorem and Lehmer’s problem. Three statements:

- (P) $\inf\{M(p) : p \in \mathbb{Z}[x], M(p) > 1\} > 1$ (Lehmer’s problem; open).
- (C) that infimum equals $M(L) = 1.17628\dots$ (Lehmer’s conjecture; open, stronger).
- (B) $\mathcal{S} \subseteq \mathfrak{B} \Rightarrow M(\mathcal{S}) \subseteq \{1\} \cup [\mu_{\mathcal{S}}, \infty)$, realised at φ (this paper; proved).

Proposition 9.1 (Logical position of the box theorem). (i) *(C) implies the floor that (B) asserts for $\mathcal{S}$; thus (B) is logically weaker than (C), and far narrower in scope ($\mathcal{S}$ versus all of $\mathbb{Z}[x]$).* (ii) *(B) does not imply (P): it says nothing about Salem numbers outside $\mathcal{S}$.* (iii) *(B) is unconditional — its proof assumes neither (P) nor (C).* (iv) *(B) is consistent with both truth values of (P).*

Proof. (i) If the global infimum exceeds 1, then *a fortiori* the infimum over the subset of emittable directions exceeds 1; so (C) (hence (P)) gives (B)’s floor immediately. The converse fails by (ii). (ii) Lehmer’s number L is a Salem number that exists independently of the construction; Theorems 5.3–5.4 place m_L *outside* the box and therefore exclude L from $\mathcal{S}$, but they assert nothing about the value of $M(L)$ or about Salem numbers in general. (iii) The proof of (B) uses only Lemmas 3.1–4.4 and the closure of the operators, none of which presupposes a global height bound. (iv) Suppose (P) is false, i.e. Salem numbers accumulate at 1. Every such small Salem number violates a wall of the box (irrational on-circle angle, by Lemma 4.4), hence is not emitted; (B) still holds. Suppose (P) is true; then (B) holds *a fortiori*. So (B) neither entails nor contradicts (P). $\square$

Corollary 9.2 (The sidestep). *The construction obtains the positive cost floor required by (2) without deciding Lehmer’s problem. It confines its own emissions to Lehmer’s Box, from which the entire Salem locus — and with it the whole open band $(1, \mu_{\mathcal{S}})$ — is provably absent. Lehmer’s number is excluded as an emission, never as a number; the open problem is left exactly where it stood.*

Remark 9.3 (What is genuinely open, stated not hidden). Two fronts remain. (a) The *free* (non-self-action) commutator at unbounded size is the one door (Lemma 8.2); it is handled per instance by the Sturm-decidable flip-straddle (Lemma 7.2) and the exact guard (Proposition 8.6), and uniformly for the framework’s actual commutator (Theorem 8.5), but not by a single uniform theorem covering *every* free commutator. [OPEN] (b) Lehmer’s problem (**P**) itself remains open in number theory, untouched. [OPEN] Neither is concealed; both are named.

10 Epistemic ledger

Each load-bearing claim, its status, and the machine-verified test that backs it. Statuses follow the companion suite’s discipline: [FORCED]/[COMPUTED] are asserted and proved here *and* re-derived in exact arithmetic; [DECLARED]/[OPEN] are modelling choices or named open fronts, recorded as context.

Claim	Status	Backing test
Empty quadratic strip $(1, \varphi)$; minimiser φ at disc 5	[FORCED]	P1-GATE-02/03
Operators preserve the floor; image $\subseteq \{1\} \cup [\varphi, \infty)$	[FORCED]	P2-EALG-01/02
Catalog arguments in $\frac{\pi}{2}\mathbb{Z}$	[FORCED]	P2-ANG-01
$\frac{\pi}{2}\mathbb{Z}$ closed under $+$ and doubling, not halving	[FORCED]	P2-ANG-02
On-circle $+$ arg in $\frac{\pi}{2}\mathbb{Z} \Rightarrow$ fourth root of unity	[FORCED]	P2-ANG-03
Salem on-circle conjugates are not roots of unity	[FORCED]	P2-ANG-04
No Salem emitted (Box exclusion)	[FORCED]	P2-ANG-05
Positive cost floor $2c \log \varphi$, linear in free c	[FORCED]	P2-GAP-03, P2-EALG-06
Salem field signature $(2, m - 1)$, Lorentzian	[FORCED]	P2-NL-02/03
27-subfield signature lattice; four $(2, 1)$ quartics	[COMPUTED]	P2-UNIF-03
Minimal degree-four Salem $\beta_4 > \varphi$; image $\cap$ sub- φ Salem $= \emptyset$	[FORCED]	P2-UNIF-04/05
Self-action difference spectrum $\{0, \pm\sqrt{5}\}$; uniform floor	[FORCED]	P2-UNIF-01
Trace-down: Salem $\iff$ flip-straddle at $t = 2$	[FORCED]	P2-DELTA-02
Circulant emits no Salem (cyclo-tomic/CM)	[FORCED]	P2-CIRC-01
Closure guard ladder; $\beta < \varphi$ by sign in $\mathbb{Q}(\sqrt{5})$	[FORCED]	P2-GUARD-01 . . 05
Conformal constant c (Jeffreys reading $c = 1$)	[DECLARED]	—
Free commutator at unbounded size; Lehmer's problem	[OPEN]	—

Closing statement. Lehmer's Box is not a new theorem so much as a new *shape* for an old one: the floor wall φ and the lattice walls $\frac{\pi}{2}\mathbb{Z}$ are the height and angle faces of a single confinement, with the field signature and the trace-down flip as its other two faces. The construction stays inside the box because every spectral operation respects both walls, and the one operation that could leave — the free commutator — is closed structurally where it is the self-action and locked operationally where it is not. Salem numbers, Lehmer's among them, live outside the box. We have not shown they cannot come close to 1; we have shown the construction never goes to fetch them. That is the precise sense in which this work sidesteps Lehmer's problem without disproving it.

References

- [1] D. H. Lehmer, *Factorization of certain cyclotomic functions*, Annals of Mathematics (2) **34** (1933), 461–479.
- [2] C. J. Smyth, *On the product of the conjugates outside the unit circle of an algebraic integer*, Bulletin of the London Mathematical Society **3** (1971), 169–175.
- [3] E. Dobrowolski, *On a question of Lehmer and the number of irreducible factors of a polynomial*, Acta Arithmetica **34** (1979), 391–401.
- [4] R. Salem, *Power series with integral coefficients*, Duke Mathematical Journal **12** (1945), 153–172.
- [5] D. Lind, K. Schmidt, T. Ward, *Mahler measure and entropy for commuting automorphisms of compact groups*, Inventiones Mathematicae **101** (1990), 593–629.